# Evidence-Grounded Flood Inundation and Infrastructure Accessibility Mapping from Sentinel-1 SAR in High-Relief Mountain Terrain

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Trishuli River Basin, Nepal · Research case study: 26 August 2026 · October 2026

Abstract. This study evaluates repeat-pass Sentinel-1 C-band synthetic aperture radar (SAR) observations for post-event surface-change screening in the Trishuli River basin, Nepal. Two Level-1 GRD observations (24 August and 5 September 2026; descending relative orbit 19, VV polarization) were processed using ESA SNAP Range-Doppler Terrain Correction and the Copernicus 30 m DEM.

A common analysis grid of 1,526.05 km² yielded 126.60 km² of candidate floodwater and 108.21 km² of candidate debris/scour under ±3.5 dB change thresholds. Held-out comparison with Copernicus EMSR927 event footprints produced 37.16% spatial coincidence in AOI03 and 23.70% in AOI01.

The workflow identified 1,062 potentially exposed buildings and 35.12 km of potentially exposed roads. The 10-day post-event delay and terrain-shadow limitations constrain interpretation. These are screening results, not field-confirmed damage or a trained predictive model.

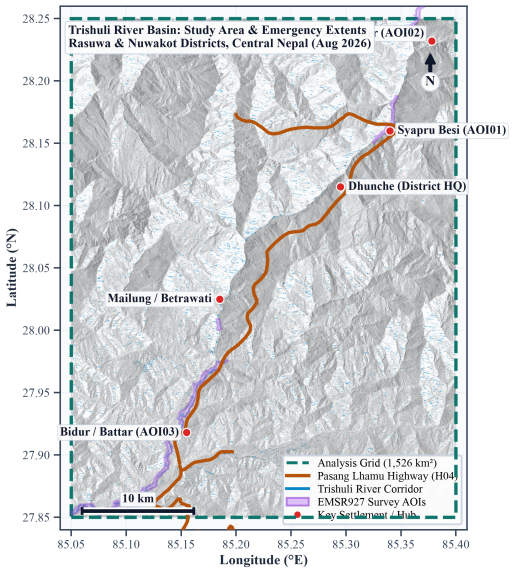



*Figure 1. Study area and reference geography. High-resolution orthorectified Sentinel-1 SAR reference geography, Trishuli River basin, Nepal.*

Keywords: Sentinel-1; synthetic aperture radar; flood mapping; change detection; Trishuli; Nepal; Copernicus EMSR927; infrastructure exposure.

## 1. Introduction

Cloud cover limits optical observations during monsoon disasters, while steep relief introduces geometric and radiometric ambiguity into SAR imagery. The 26 August 2026 Trishuli River flood provides a case for testing whether repeat-pass radar anomalies can identify persistent surface change and support infrastructure screening. The analysis asks where significant backscatter changes persist, how these changes intersect mapped assets, and how the resulting footprint corresponds to independently mapped Copernicus emergency observations [1–3].

## 2. Study Area and Source Data

The analysis domain spans approximately 85.05–85.40° E and 27.85–28.25° N across Rasuwa and Nuwakot districts, Nepal. Relief ranges from river terraces to high Himalayan slopes. The reported common analysis grid covers 1,526.05 km². Dataset provenance is summarized in Table 1; exact product identifiers and source artifacts should accompany the reproducibility archive.

*Table 1. Observation and reference datasets.*

| Dataset / source | Acquisition or role | Specification |
|---|---|---|
| Sentinel-1 GRD, CDSE [1] | Pre: 24 Aug 2026, 00:18:44 UTC | IW VV, relative orbit 19, descending |
| Sentinel-1 GRD, CDSE [1] | Post: 05 Sep 2026, 00:18:44 UTC | IW VV, relative orbit 19; 12-day repeat |
| Copernicus DEM [2] | Range-Doppler terrain correction | Global 30 m DEM |
| Copernicus EMSR927 [3] | Independent reference | AOI01 and AOI03 vector products |
| OSM / Nepal Survey Department [4,5] | Building and road exposure | Vector networks and footprints |

## 3. Methods

### 3.1. Radar preprocessing and radiometric interpretation

Matching-orbit Sentinel-1 GRD acquisitions were processed using ESA SNAP with orbit application, calibration, noise handling and Range-Doppler Terrain Correction against Copernicus 30 m elevation data. Processing notes document an SNAP warning concerning the calibration lookup table in COG-wrapped SAFE metadata. Internal IPF 004.03 vector inspection is not independent proof of absolute radiometric accuracy. The exported single-band GeoTIFF did not preserve the auxiliary layover/shadow mask, limiting interpretation in steep terrain.

### 3.2. Bitemporal change classification

For each common-grid pixel, backscatter in decibels was calculated as $10 \log_{10}(\sigma^\circ)$, with $\Delta\sigma^\circ = \sigma^\circ\text{post} - \sigma^\circ\text{pre}$. Candidate inundation was assigned when $\Delta\sigma^\circ \leq -3.5$ dB; candidate debris/scour when $\Delta\sigma^\circ \geq +3.5$ dB. Pixels with both dates $\leq -16$ dB were treated as persistent water. These are physically motivated threshold classes, not predictions from a trained machine-learning classifier. Terrain speckle, vegetation change, shadow and residual calibration differences may cause false positives.

### 3.3. Exposure and independent reference comparison

Candidate hazard layers were spatially intersected with mapped buildings and road segments. Counts and lengths represent unique intersections, not confirmations of structural failure or traffic closure. EMSR927 optical disaster polygons were reserved for independent footprint comparison. Because the optical reference was acquired near the event while post-event SAR was acquired 10 days later, overlap percentages are not conventional classifier accuracy, precision or recall.

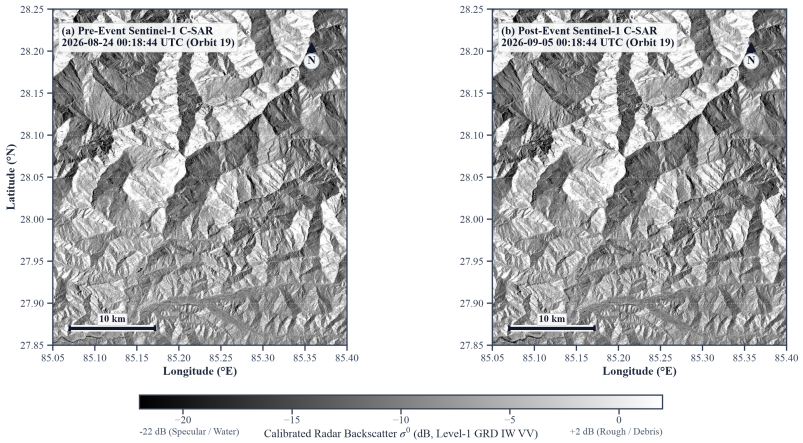



*Figure 2. Pre-event (24 August) and post-event (5 September) Sentinel-1 radar intensity comparison processed on common analysis grid using identical radiometric scale.*

## 4. Results

*Table 2. Raw common-grid threshold classification, before morphology filtering.*

| Class | Decision rule | Pixels | Area (km²) | Share |
|---|---|---|---|---|
| Candidate inundation | $\Delta\sigma^\circ \leq -3.5$ dB | 1,439,895 | 126.60 | 8.30% |
| Candidate debris/scour | $\Delta\sigma^\circ \geq +3.5$ dB | 1,230,733 | 108.21 | 7.09% |
| Persistent water | Both dates $\leq -16$ dB | 755,548 | 66.40 | 4.35% |
| Unchanged | $-3.5 < \Delta\sigma^\circ < +3.5$ dB | 13,931,062 | 1,224.84 | 80.26% |
| Total | Common analysis grid | 17,357,238 | 1,526.05 | 100% |

The source analysis reports geodesic verification of candidate floodwater area: 126.6146 km² by row-wise WGS84 integration and 126.5830 km² by metric reprojection, compared with 126.60 km² using centroid pixel area. This agreement tests areal arithmetic, not thematic detection accuracy.

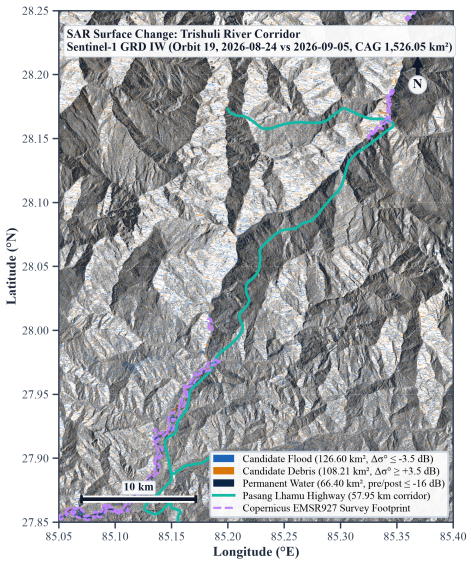



*Figure 3. Candidate SAR surface-change classes across the study domain. Bitemporal classification over 1,526.05 sq km common analysis grid.*

### 4.1. Contiguous hazard morphology

The dashboard applies a minimum mapping unit of 1,000 m² (10 connected 10 m-classification pixels) to suppress isolated speckle. Filtered outputs are 62.86 km² of contiguous candidate floodwater and 46.93 km² of contiguous candidate debris/scour. These must be distinguished from the raw totals in Table 2.

## 5. Copernicus EMSR927 Reference Comparison

Table 3 reports spatial coincidence with optical event polygons from EMSR927. The observations are not temporally equivalent; the measurements therefore do not establish recall or precision.

*Table 3. Held-out optical-footprint coincidence by sector.*

| Sector | Optical area | Flood overlap | Debris overlap | Combined |
|---|---|---|---|---|
| AOI03 Bidur | 5.89 km² | 1.58 km² (26.89%) | 0.61 km² (10.28%) | 37.16% |
| AOI01 Syapru Besi | 1.11 km² | 0.11 km² (9.82%) | 0.15 km² (13.88%) | 23.70% |

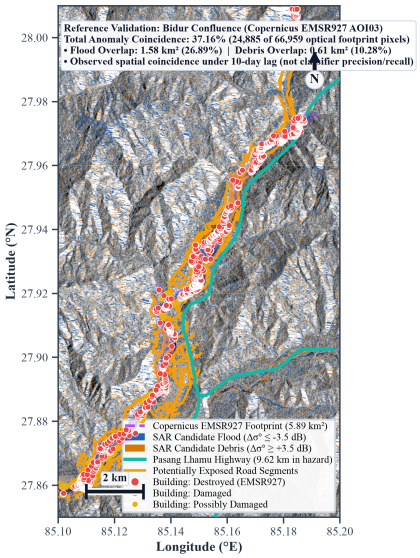



*Figure 4. EMSR927 event footprint and SAR anomaly overlay at Bidur confluence (AOI03 validation sector, 37.16% spatial coincidence).*

## 6. Infrastructure Exposure and Access

Spatial intersection identified 1,062 potentially exposed buildings of 3,816 surveyed (27.83%) and 35.12 km of potentially exposed roads of 260.71 km (13.47%). Of 2,813 structures graded “Destroyed” in EMSR927, 936 intersect SAR anomaly pixels. This does not establish that SAR independently detected their physical condition.

*Table 4. Building exposure cross-tabulation against EMSR927 grades.*

| Reference grade | Flood | Debris | Unique exposed | Surveyed | Share |
|---|---|---|---|---|---|
| Destroyed | 812 | 124 | 936 | 2,813 | 33.27% |
| Damaged | 48 | 21 | 69 | 354 | 19.49% |
| Possibly damaged | 24 | 33 | 57 | 649 | 8.78% |
| Total | 884 | 178 | 1,062 | 3,816 | 27.83% |

For the Pasang Lhamu Highway corridor, 9.62 km of 57.95 km intersected candidate hazard zones (16.59%). Routing experiments treated these intersections as hypothetical closures and flagged six candidate settlements as potentially disconnected from hospital access. These depend on closure assumptions, not verified road impassability.

## 7. Limitations and Scientific Interpretation

Observation timing. The post-event pass was 10 days after the reported event; persistent radar anomalies cannot reconstruct maximum flood extent. Topographic ambiguity. The exported product lacks an operational terrain-shadow mask, and dark slopes may mimic water. Radiometry. The COG-related SNAP LUT warning requires disclosure; metadata inspection is not independent calibration. Attribution. Positive backscatter change can reflect roughening, deposition, land cover or acquisition effects. Exposure. Intersections do not prove destruction, closure or inaccessibility. Validation. Temporally offset optical and radar observations support spatial-coincidence analysis only.

## 8. Reproducibility and Data Availability

The VARUNA experiment archive reports a three-band flood-change GeoTIFF, candidate hazard GeoJSON, spatially indexed FlatGeobuf, independent EMSR927 validation JSON, and infrastructure-exposure JSON. Exact file hashes, processing-graph revision and software versions should accompany any public release. No training split, learned weights or training dataset are reported because this analysis uses deterministic thresholds.

## 9. Conclusion

VARUNA demonstrates an auditable radar-based disaster-screening workflow for steep terrain. Outputs include candidate hazard footprints, potential asset exposure and explicitly time-qualified correspondence with independent event features. Operational use requires stronger shadow handling, acquisitions closer to the flood peak, and ground verification before labeling infrastructure damaged or routes closed.

## References

[1] European Space Agency / Copernicus Data Space Ecosystem. Sentinel-1 Level-1 GRD products and technical documentation (2026).

[2] European Space Agency / European Union. Copernicus DEM Global 30 m, product documentation.

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[6] Bonafilia, D., et al. (2020). Sen1Floods11: A Georeferenced Dataset to Train and Test Deep Learning Flood Algorithms for Sentinel-1. CVPR Workshops.

[7] Small, D. (2011). Flattening Gamma: Radiometric Terrain Correction for SAR Imagery. IEEE Transactions on Geoscience and Remote Sensing, 49(8), 3081–3093.

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